HyperMLL: Toward Robust Hyperspectral Image Classification With Multisource Label Learning

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***Abstract*— Recent advancements in hyperspectral image (HSI) classification rely on deep learning, demanding high-quality labels. Traditional methods of label collection are costly and expertise-intensive. Alternative approaches, such as crowdsourcing, large-scale models, and others, have emerged as cost-effective solutions for label acquisition. However, labels obtained from these sources often exhibit noise. Currently, there is a research gap in addressing the challenges posed by multisource noisy labels in HSI interpretation. To bridge this gap, we introduce a novel method for effectively interpreting HSI with multisource label learning (HyperMLL). First, we develop a novel multisource label noise simulation approach for HSI based on spectral features. Then, we propose the spectral- and spatial-related confusion matrices (CMs) to learn the distribution of noisy labels, aiming at mitigating the impact of multisource noisy labels. In addition, these learned CMs are leveraged to assess the overall credibility and class-specific credibility of different label sources, laying the foundation for future work to ensure label quality while reducing labeling costs. We generate multisource noisy labels at different proficiency levels and conduct detailed experiments on three publicly available datasets. The experimental results demonstrate that our framework outperforms state-of-the-art HSI noise learning methods by a large margin and can accurately evaluate the sourcewise label qualities.**

***Index Terms*— Deep learning, hyperspectral image (HSI) classification, multisource labels, noisy labels.**

I. Introduction

HYPERSPECTRAL remote sensing images have been widely applied in various fields, such as smart agriculture [1], environmental monitoring [2], mineral exploration [3], urban planning [4], and so on. A hyperspectral image (HSI) is primarily used in smart agriculture for crop monitoring, pest and disease identification, and soil analysis. By capturing the spectral information of plants, it can accurately determine the health status, nutritional needs, and growth conditions of crops, thereby enabling refined agricultural management and improving the yield and quality of crops. In environmental monitoring, HSI can be used for atmospheric monitoring, water quality monitoring, and land cover change analysis. It can identify and quantify environmental pollutants, monitor ecosystem health, and provide high-precision environmental data, contributing to environmental protection and sustainable development. In mineral exploration, HSI is used for mineral identification, ore deposit localization, and geological analysis. Its hyperspectral data can identify the spectral characteristics of surface minerals, providing information on mineral distribution and types, aiding geologists in resource assessment and mining planning. In urban planning, HSI is used for land use classification, urban green space monitoring, and building analysis. It can provide rich surface information, helping urban planners make informed decisions, optimize urban layouts, and enhance urban management. However, achieving high-precision interpretation of remote sensing images requires high-quality annotations, which typically involve time-consuming and labor-intensive processes of field measurement and pixelwise analysis by professionals, making it difficult to scale up [5], [6]. Therefore, research on how to utilize low-quality labels to train models for obtaining high-precision interpretation results is crucial. This is also the problem that hyperspectral noise learning aims to address.

At present, hyperspectral noise learning primarily addresses scenarios where each sample has only one label [7], [8], in which these labels exhibit a certain degree of noise, meaning some labels are correct, while others are wrong. Researchers typically train models by selecting high-quality labels to achieve higher classification accuracy [9], [10]. However, such approach has some limitations. First, obtaining labels of hyperspectral data is restricted, requiring field surveys and expert knowledge for annotation, making it challenging to expand the size of the labels [5]. Second, the difficulty of labeling varies among different classes, with some classes more prone to labeling errors, and in some cases, all samples of certain classes may be mislabeled, resulting in lower classification accuracy for these classes. In addition, lacking source information for labels makes it difficult for the model to differentiate between reliable and

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noisy labels, possibly resulting in misleading outcomes during model training. Therefore, we believe that the setting of single-label-per-sample is not suitable for the practical situation of hyperspectral noisy label learning. Furthermore, compared with other remote sensing data, hyperspectral data interpretation relies more on expert knowledge, thus requiring the preservation of source information for labels to fully exploit the potential of each label. Thus, enhancing hyperspectral multisource noise learning requires consideration of intricate noisy label models and noise suppression strategies to enhance model performance and generalization capabilities.

To this end, we propose a more realistic setting for hyperspectral noise learning, where labels for hyperspectral data can originate from multiple sources, such as field measurements, remote sensing images of different resolutions, historical data, and generated from large-scale models. Hence, each sample can possess labels from multiple sources. We assume all label sources annotate every pixel in the HSI, encompassing both correct and erroneous labels. With multiple sources and labels, we aim to achieve two objectives: 1) to train highly accurate classification models capable of providing precise classifications for HSI across diverse applications and 2) to evaluate the quality of different label sources, which is vital for model training and label collection. For instance, based on the evaluation of different sources, we can strike a balance between the cost and quantity of labels, identify categories prone to wrong classification, and assess the quality of labels from different sources, thus guiding subsequent label collection and model training endeavors.

Previous HSI noise learning methods failed to yield satisfactory results in the this realistic setting for three reasons.

1) The existing noise simulation methods primarily employ percentage-based simulation [7], [8], [11], where correct labels are modified to incorrect ones based on preset ratios. This is a common strategy in simulating noise for natural images, while hyperspectral data exhibit more pronounced spectral variability due to phenomena such as spectral variability within the same object and spectral ambiguity between different objects. These characteristics are crucial in generating hyperspectral noise annotations that better reflected real-world data collection scenarios. The limitations of previous hyperspectral noise simulation methods hinder their applicability in the real world.
2) Previous hyperspectral noise learning methods overlook the prior knowledge of different label sources [5]. In reality, labels from different sources exhibit unique characteristics and biases. Incorporating this prior knowledge could enhance the accuracy of hyperspectral noise learning and improve model robustness. However, the existing methods have not effectively utilized this prior knowledge, limiting the performance of the models.
3) These methods assumed one sample corresponds to one label [10], making them incompatible with multilabel datasets. When dealing with labels from multiple sources, such as three annotation sources, the input training samples would triple, leading to a significant increase in computational cost. This adversely affected the efficiency of these methods in handling larger scale datasets or training models with limited resources.

Therefore, this article explores intelligent interpretation of HSI with multisource noisy labels. Our main contributions are as follows.

1) We propose a novel noisy label simulation method that better suits hyperspectral remote sensing data, providing new perspectives for future research in hyperspectral noise learning.
2) We present a class-specific confusion matrix learning algorithm that leverages both spectral and spatial deep features. It effectively exploits noisy labels from various sources, maximizing their utility.
3) We establish a comprehensive HSI multisource noise learning framework and propose a way to assess label qualities of different sources. To the best of our knowledge, this work presents the first attempt in multisource HSI noise learning.

II. Related Works

A. HSI Classification

HSI interpretation involves categorizing and identifying various objects or targets within an image acquired through hyperspectral technology. In HSI, each pixel contains a wealth of spectral information, representing the reflectance or radiance characteristics at different wavelengths. These features can be used to distinguish between different objects, features, or environments on the ground. The goal of HSI interpretation is to analyze these spectral features and utilize various machine learning or deep learning algorithms to classify the pixels in the image according to the types of features they represent, thereby achieving automated identification and classification of objects. We classify commonly employed methods for HSI interpretation into three categories: traditional machine learning, convolutional neural network (CNN)-based classification approaches, and attention mechanism-based methods exemplified by Transformers.

Due to the typically high-dimensional nature of hyperspectral data, SVM methods have become commonly used in traditional HSI interpretation. SVM exhibits good robustness when dealing with high-dimensional data and can effectively exploit both spectral and spatial features of HSI. For example, [12] argued that combining multiple features (e.g., spectral and shape) is key for HSI representation, and they proposed a patch-based linear combination method to find the optimal solution, while [13] emphasized the importance of the kernel, which is used in this study to extract spatial and spectral features from HSI. By comparing the similarity between these features, different land cover classes can be distinguished. In [14], SVM was employed for pixel-level classification of HSI, followed by the utilization of Markov random field regularization to refine the obtained classification results, thus further enhancing the accuracy of classification.

CNNs are a commonly used deep learning method extensively applied in the field of image processing. Through

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structures, such as convolutional layers, pooling layers, and fully connected layers, CNNs can automatically learn features within images. The 3-D nature of HSI makes CNNs particularly prominent in jointly exploiting spectral and spatial information [15], [16]. Zhong et al. [17] proposed a method that directly handles raw 3-D data cubes by introducing residual blocks and batch normalization, thereby improving the accuracy of HSI classification. Zhang et al. [18] integrated joint representations across multiple regions to effectively utilize spectral details, spatial structure, and contextual semantics for pixel-level analysis.

The Transformer is a deep learning architecture designed for sequence-to-sequence tasks [19]. Built upon a unique self-attention mechanism, it allows models to dynamically establish relationships between different positions in a sequence. This characteristic enables the Transformer to better exploit and represent the sequential properties of spectral features in HSI classification compared with CNNs [20], [21], [22]. Huang et al. [23] proposed a novel approach to address the challenge of training Transformer-based HSI classifiers with limited training samples, developing three methods to facilitate discriminative feature learning and integrate them into the approach. Xu et al. [24] introduced a novel network aimed at exploring diverse local structures of various configurations within HSI, comprehensively capturing both global and local multiscale pixel features for effective classification. Meanwhile, [25] proposed a framework aimed at mitigating the problem of incorporating irrelevant spatial information during the extraction of spectral–spatial (SS) features.

B. Noisy Learning

Current deep learning methods for HSI classification generally assume that labels are clean, meaning they do not consider noise issues and often lack attention to the problem of multisource noise. However, noisy labels are prevalent in reality, such as labeling uncertainty or errors. Therefore, addressing label noise in HSI classification is an important issue. Neglecting noise in HSI classification research can result in adverse outcomes, including reduced prediction accuracy, heightened model complexity, and an increased demand for training samples in inference models [26]. Some researchers proposed using noise filtering methods to address the problem of noisy labels [9], [27], selecting clean and reliable labels from noisy labels to train models. Others attempted to improve HSI classification accuracy by correcting noisy labels and fully utilizing all label information for model training [28].

However, the noise learning methods mentioned above are primarily applicable to natural images or medical images and may not be suitable for HSI noise learning due to the high dimensionality and spatial–spectral complexity of hyperspectral data. Because of factors, such as the difficulty and high cost of annotation, the problem of noisy labels in HSI is prevalent. How to utilize noisy labels to ensure the accuracy of HSI classification has become a challenge that attracts considerable attention. Consequently, researchers primarily attempt to address this challenge through noise filtering methods, data augmentation, and noise learning methods.

In the noise filtering methods, [10] proposed a novel framework that integrates density peak clustering with distance weighting between super-pixels and pixels, effectively identifying and eliminating noisy labels in the training dataset prior to HSI classification. Jiang et al. [29] introduced a label noise cleaning method utilizing a correlation graph constructed from the similarities in both spectral and spatial characteristics of HSI. This method utilizes spectral feature vectors to measure similarity and regulates the label noise cleaning process accordingly. Gong et al. [30] introduced an innovative deep learning framework designed for HSI classification. This approach incorporates a CNN network featuring a noise skew module and a denoising framework aimed at improving classification accuracy. Removing noisy labels reduces the total number of labels available for HSI, thereby limiting improvements in classification accuracy. In contrast, noise augmentation methods effectively leverage valuable information from noisy labels and facilitate easier acquisition of these labels [8]. Noise learning methods integrate the advantages of noise filtering and noise augmentation, enabling more effective utilization of noisy labels. Wang et al. [31] employed a method that adjusted parameters to manage the trade-off between different measures of entropy, aiming to mitigate the effects of noisy labels. Roy et al. [32] tackled HSI classification with noisy labels by employing a method that combined two types of convolutional kernels (spectral and spatial) to produce final feature maps. This approach was designed to mitigate sensitivity to noise and computational complexity, thereby addressing challenges stemming from inaccurate training label. Zhang et al. [33] introduced a dual-network framework for unified label noise learning. They devised a methodology based on consistency and inconsistency between networks, employing a joint loss function that combines supervised losses from both networks and evaluates their relative differences to enhance classification accuracy. Zhang et al. [7] introduced a novel framework for HSI noise learning, which utilized clustering-level, instance-level, and structure-level representations through defined triple learning losses. This approach successfully alleviated the issue of classifier overfitting to noisy labels.

Current methods primarily consider single-source noisy labels in HSI. However, noisy labels can originate from multiple sources, meaning a hyperspectral pixel may have different labeling distributions. These multisource noises can stem from various factors, such as different annotators, historical annotations, and annotations from different modalities. The existing research overlooks these complex and more realistic scenarios of noisy labels, resulting in limitations in the robustness, accuracy, and efficiency of models in practical applications. Therefore, studying multisource noisy labels for HSI classification is a closer-to-reality yet overlooked important issue.

C. Learning With Multisourced Labels

In domains, such as natural image processing and medical image classification, researchers have made some explorations in learning from multisource labels [34], [35]. Due to the

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high cost of expert annotations, nonexpert labeling has become an effective means of obtaining a large number of labels, among which crowdsourcing not only offered cost-effectiveness but also enabled rapid acquisition of abundant labeling data [36], [37]. However, the accuracy of labels from nonexpert annotators was questionable, and label noise significantly impacted the classification accuracy of models. Therefore, many researchers attempted to train network models using noisy labels from multiple annotators and made certain progress [38].

In natural image classification, [39] proposed a method to improve classification accuracy by constructing a model that correlated annotator expertise with image classification difficulty. They also introduced a new form of annotation called soft annotation, using it as ground truth (GT) instead of the original noisy labels during training to reduce the impact of label noise on classification accuracy. Gao et al. [40] aimed at improving classification accuracy by learning annotator weight vectors and confusion matrices (CMs) in scenarios where each data point received multiple labels from multiple annotators. Chu et al. [41] introduced a crowdsourcing-based classification model for handling noisy labeling, which differentiated between noise patterns and facilitated learning of shared and individual CMs. This approach aimed to enhance classification accuracy by effectively modeling noise distributions. Lin et al. [42] introduced a crowdsourcing-based active learning approach that encouraged annotators to focus on labeling labels they were confident in, thus reducing labeling errors and improving classification accuracy while reducing costs.

In medical image classification, [43] introduced deep Gaussian processes to effectively manage multisource labels, thereby substantially enhancing the classification accuracy of medical images. Guan et al. [44] proposed modeling each annotator individually and assigning greater weights to more reliable annotators. This strategy not only improved the accuracy of medical image classification but also capitalized on the strengths of individual experts in specific types of data classification. In remote sensing image classification, [45] introduced a majority voting (MV) method to construct benchmark datasets from labels provided by multiple annotators. They developed a multiclass classification framework applicable to forest ecosystems using both LiDAR and raster images. Wang et al. [46] argued that remote sensing data often involved numerous categories, which were prone to similarities that made them difficult to distinguish. Nonexpert annotators might make errors in these indistinguishable categories, leading to erroneous MV labels. In such cases, the MV method might not be effective. Therefore, they proposed the manual design of an annotator competence assessment process aimed at assigning higher weights to annotators with stronger capabilities than those with weaker capabilities, thus reducing the impact of noisy labels on classification accuracy.

However, most of the aforementioned multisource noise learning approaches focus on normal images and medical images, all of which are not applicable to HSI classification. HSI contains spectral data in hundreds of bands, involving not only high-dimensional information acquisition and processing but also a higher likelihood of interclass confusion due to the potentially very similar spectral characteristics of different materials across different bands. Furthermore, the existing noisy label simulation methods and processing algorithms fail to fully utilize the information in hyperspectral data, resulting in biases between the simulated process and the actual situation, leading to a decrease in model classification performance. Therefore, establishing an effective multisource noise learning framework for HSI classification is an important problem that needs to be addressed.

III. Method

In this study, we propose a new framework named HyperMLL to address the problem of HSI classification with multisource noisy labels. The overview of HyperMLL is shown in Fig. 1. It comprises three key components: the simulation of multisource noisy labels in Section III-A, the training of CMs for different label sources in Section III-B, and the credibility evaluation of these label sources in Section III-C. In Fig. 1, the first row shows how multisource noisy labels are simulated according to spectral similarities between HSI patches, the second row describes the learning of CMs, and the last row illustrates the estimations of classwise label credibility for each source.

The HSI set is represented as $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^{N}$, where N denotes the number of pixels and $\mathbf{x}$ is the spectral vector of the ith pixel. $\mathbf{X}$ has a size of $W \times H \times C$, where W is the width, H is the height, and C is the number of spectral channels. Given a patch $\mathbf{h}_i$ with size d centered on a pixel $\mathbf{x}_i$, the collection of all pixelwise patches is represented as $\mathcal{H}$. The set of noisy labels is represented as $\{\tilde{y}_i^r\}_{i=1,\cdots,N}^{r=1,\cdots,R}$, where $\tilde{y}_i^r$ represents the label given by a label source r to pixel i and R denotes the number of label sources. The GT labels are represented as $\{y_i\}_{i=1}^{N}$, where y_i represents the true label of pixel i, $y_i \in Z$, and $Z = \{z_k\}_{k=1}^{K}$ denotes a set of land cover classes, where K represents the number of land cover classes. It should be noted that, in this context, we only consider the closed set classification problem [47].

A. Multisource Label Generation Method

When simulating noise, we take into account the following two factors: 1) the phenomenon where different land cover classes exhibit very similar spectral curves, known as spectral confusion, often leads to mislabeling during manual classification and 2) since the labels we use come from multiple sources, the level of expertise in labeling varies among different sources, meaning that the accuracy of labels from different sources varies. The existing methods for simulating noisy labels in HSI mainly involve flipping class based on percentages, without considering the characteristics of hyperspectral data. Therefore, we propose a new and more realistic method for simulating HSI noisy labels from multiple sources.

For label source r, we randomly select a land cover class as their error-prone category z_k. Within this category, we randomly choose a spectral vector of a pixel from all spectral vectors of class z_k, as the spectral vector x_i^* of this

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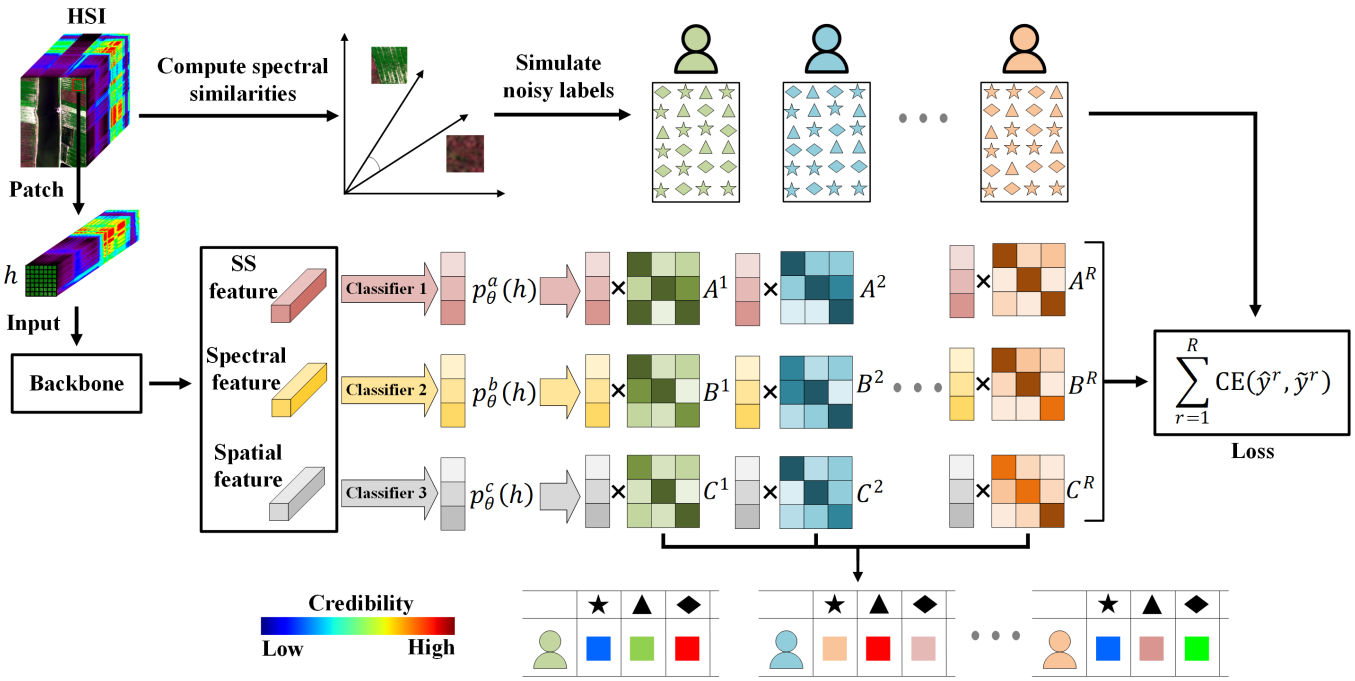

Fig. 1. Framework of HyperMLL. We consider an HSI with three types (☆ △ ◇) of land cover classes. Initially, we simulate multisource noisy labels $\{\tilde{y}^r\}_{r=1}^R$ by introducing spectral similarity between spectral vectors, as shown in the first row. Then, an HSI patch $\mathbf{h}$ is fed into a backbone network to extract SS features, spectral features, and spatial features and obtain corresponding predictions $\mathbf{p}_\theta^a(\mathbf{h})$, $\mathbf{p}_\theta^b(\mathbf{h})$, and $\mathbf{p}_\theta^c(\mathbf{h})$, respectively. These results are then multiplied by their associated CMs and summed to produce the weighted model predictions $\{\hat{y}^r\}_{r=1}^R$ for the computation of the cross-entropy loss, as shown in the middle row. Furthermore, utilizing the learned CMs, we can compute the class-specific credibility and overall credibility for different label sources, as shown in the last row.

error-prone category, where $i \in \{y_i = z_k\}$. Then, we compute the spectral similarity between all pixel spectral vectors and x_i^* using cosine similarity calculation method, as shown in the following equation:

$$s_j = \arccos\left(\frac{x_i^* \cdot x_j}{\|x_i^*\| \times \|x_j\|}\right) \tag{1}$$

where s_j represents the spectral similarity between the spectral vector x_j of pixel j and x_i^*. To provide a clearer representation of the spectral similarity between two pixels, we quantify it using the angle between their respective spectra. Considering the factors mentioned above, we formulate the simulation label generation for annotator r, as depicted in the following equation:

$$\tilde{y}_i^r = \begin{cases} z_k, & \text{if } s_j \le l_r \\ y_i, & \text{others} \end{cases} \tag{2}$$

where l_r represents the upper bound of label source r's proficiency level. If $s_j \le l_r$, label source r would label pixel j as z_k; otherwise, label pixel j as the GT y_i. Different label sources have different upper bounds of proficiency levels, and the upper bounds set of proficiency levels for all label sources is denoted by L, where $L = \{l_r\}_{r=1}^R$. A toy example for three label sources is given in the first row in Fig. 1.

B. SS Confusion Matrix Learning

After padding pixel i to obtain the matrix $\mathbf{h}_i$ and feeding it into the network backbone, we get SS fusion features, spectral features, and spatial features. Then, send these features into the classifier, and we get classification results from SS fusion feature $\mathbf{p}_\theta^a(i)$, spectral feature $\mathbf{p}_\theta^b(i)$, and spatial feature classification results $\mathbf{p}_\theta^c(i)$. We construct CMs of size $K \times K$ for the three classification results, denoted as $\mathbf{A}^r = (a_{z_i z_j}^r)$, $\mathbf{B}^r = (b_{z_i z_j}^r)$, and $\mathbf{C}^r = (c_{z_i z_j}^r)$. Taking $a_{z_i z_j}^r$ as an example, it represents the probability that label r labels class z_i as class z_j, serving to correct the spectral feature classification results and mitigate the impact of misclassification on the network. $b_{z_i z_j}^r$ and $c_{z_i z_j}^r$ follow the same logic. We aim not only to obtain the probability distribution of pixelwise classification results $p(y \mid \mathbf{x})$ but also to learn the three CMs of all label sources.

Inspired by [11], and assuming that different label sources are statistically independent and the label noise of each pixel are independent, the probability of observing noisy labels $\{\tilde{y}^1, \cdots, \tilde{y}^R\}$ for pixel x is given by the following equation:

$$p\left(\tilde{y}^1, \cdots, \tilde{y}^R \mid \mathbf{x}\right) = \prod_{r=1}^{R} \sum_{y=z_1}^{z_K} p\left(\tilde{y}^r \mid y, \mathbf{x}\right) p(y \mid \mathbf{x}) \tag{3}$$

where $p(y \mid \mathbf{x})$ represents the probability distribution of the true label in pixel x and $p(\tilde{y}^r \mid y, \mathbf{x})$ represents the probability distribution of label source r labeling pixel $\mathbf{x}$ as $\tilde{y}^r$. Furthermore, because we assume that the label noise is independent of the input pixel, the probability of label source r labeling the true value $y = z_i$ to $\tilde{y}^r = z_j$ is independent of pixel $\mathbf{x}$, i.e., $p(\tilde{y}^r = z_j \mid y = z_i) = \alpha a_{z_i z_j}^r + \beta b_{z_i z_j}^r + \gamma c_{z_i z_j}^r$, where α, β, and γ are weighting coefficients, $\alpha, \beta, \gamma \in [0, 1]$; substituting into (3), the probability distribution of noisy labels is given by the following equation:

$$p\left(\tilde{y}^1, \cdots, \tilde{y}^R \mid \mathbf{x}\right) = \prod_{r=1}^{R} \sum_{y=z_1}^{z_K} p\left(\tilde{y}^r \mid y, \mathbf{x}\right) p(y \mid \mathbf{x}) \tag{4}$$

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Given the set of padded HSI pixels $\mathcal{H} = \{\mathbf{h}_i\}_{i=1}^{N}$ and the noisy labels $\tilde{\mathbf{Y}}^r = \{\tilde{y}_i^r\}_{i=1,\cdots,N}^{r=1,\cdots,R}$, we optimize the parameters $\{\theta, \mathbf{A}^r, \mathbf{B}^r, \mathbf{C}^r\}$ by minimizing the negative log-likelihood (NLL), i.e., $-\log p(\tilde{\mathbf{Y}}^1, \cdots, \tilde{\mathbf{Y}}^R \mid \mathcal{H})$. According to (4), minimizing the negative NLL is equivalent to computing the cross entropy between the observed noisy labels $\tilde{y}_i^r$ and the learned noisy labels distribution $\hat{y}_i^r$, as shown in the following equation:

$$-\log p\left(\tilde{\mathbf{Y}}^1, \cdots, \tilde{\mathbf{Y}}^R \mid \mathcal{H}\right) = \sum_{i=1}^{N}\sum_{r=1}^{R} CE\left(\hat{y}_i^r, \tilde{y}_i^r\right) \tag{5}$$

where the calculation of $\hat{y}_i^r$ is shown in the following equation:

$$\hat{y}_i^r = \alpha \mathbf{A}^r \mathbf{p}_\theta^a(\mathbf{h}_i) + \beta \mathbf{B}^r \mathbf{p}_\theta^b(\mathbf{h}_i) + \gamma \mathbf{C}^r \mathbf{p}_\theta^c(\mathbf{h}_i). \tag{6}$$

Finally, the loss function of the model is shown in the following equation:

$$\mathcal{L} = \sum_{i=1}^{N}\sum_{r=1}^{R} CE\left(\hat{y}_i^r, \tilde{y}_i^r\right). \tag{7}$$

C. Label Quality Evaluation Method

After learning the three CMs for each labeling source, we can assess the reliability of different label sources. We refer to the comprehensive evaluation of label sources' ability to distinguish all land cover classes as the overall credibility of label sources, calculated as shown in the following equation:

$$t^r = \frac{\alpha \mathrm{tr}(\mathbf{A}^r) + \beta \mathrm{tr}(\mathbf{B}^r) + \gamma \mathrm{tr}(\mathbf{C}^r)}{\mathrm{sum}(\mathbf{A}^r) + \mathrm{sum}(\mathbf{B}^r) + \mathrm{sum}(\mathbf{C}^r)} \tag{8}$$

where t^r represents the overall credibility of label source r; $\mathrm{tr}(\mathbf{A}^r)$ denotes the trace of the confusion matrix $\mathbf{A}^r$; $\mathrm{sum}(\mathbf{A}^r)$ represents the sum of all elements in the confusion matrix $\mathbf{A}^r$; and α, β, and γ are weighting coefficients, $\alpha, \beta, \gamma \in [0, 1]$. A higher overall credibility indicates a stronger ability of the label source to distinguish all land cover classes in HSI. In addition, through the CMs, we can also understand the ability of different label sources to distinguish a particular land cover class, which we refer to as the label source's class-specific credibility. It is calculated, as shown in the following equation:

$$t_i^r = \frac{\alpha a_{z_i z_i}^r + \beta b_{z_i z_i}^r + \gamma c_{z_i z_i}^r}{\sum_{j=1}^{N} a_{z_i z_j}^r + \sum_{j=1}^{N} b_{z_i z_j}^r + \sum_{j=1}^{N} c_{z_i z_j}^r} \tag{9}$$

where t_i^r represents the class-specific credibility of label source r for class z_i. A higher class-specific credibility indicates a stronger ability of the label source to distinguish individual land cover classes in HSI. We denote the set of overall credibility and class-specific credibility for all annotators as T, where $T = \{t^r, t_i^r\}_{i=1,\cdots,N}^{r=1,\cdots,R}$. Based on label sources' credibility, we can optimize the allocation of label tasks in subsequent work to reduce label noise while lowering label costs. This optimization aims to improve the quality of labels while reducing the cost of data collection.

A demo for class-specific and overall credibilities for different label sources is given on the last row in Fig. 1. As shown in the color bar, which ranges from blue to red indicating label credibility from low to high, the first label source denoted by green cartoon character exhibits low credibility for class 1 (☆), moderate credibility for class 2 (△), and the highest credibility for class 3 (◇).

IV. Experiments

This section will provide a detailed overview of experiments, including dataset description, experimental parameter settings, comparative methods, experimental results and analysis, and ablation studies.

A. Datasets

1) *Pavia University (PU) Dataset:* The dataset was obtained from a flight campaign in northern Italy, with the dimensions of 610 × 340 × 103. It comprises 42 776 pixels with GT labels, covering nine land cover classes and 103 spectral bands.
2) *Kennedy Space Center (KSC) Dataset:* The dataset was collected over KSC in Florida, with a spatial resolution of 18 m. It has the dimensions of $512 \times 614 \times 176$, comprising 5211 pixels with GT labels across 176 spectral bands, encompassing 13 different land cover categories.
3) *WHU-Hi-LongKou (WHLK) Dataset [48], [49]:* This high-spatial-resolution HSI (referred to as H^2) was acquired in 2018 over Longkou Town, Hubei Province, China. It possesses a spatial resolution of 0.463 m and consists of 270 spectral bands ranging from 400 to 1000 nm. The image dimensions are 550 × 400 pixels, encompassing nine land cover categories.

We select the PU, KSC, and WHLK datasets not only because they are widely used in the field of HSI classification [7], [8], [25], but also because they represent different spatial resolution characteristics. Specifically, PU and KSC are traditional datasets with low spatial resolution, while WHLK is a dataset with high spatial resolution. We believe that the selection of these datasets helps demonstrate the generality and applicability of our method. By using datasets with different spatial resolutions, we can test and validate our proposed method under various conditions. The detailed data for the three datasets are provided in Table I.

B. Implementation Details

The model was trained on a workstation consisting of a 13th Gen Intel[1] Core[2] i7-13700KF CPU with 3.4 GHz and 16 cores, along with 64 GB of memory, and a NVIDIA GeForce RTX 3090 GPU equipped with 24 GiB of memory. All the codes used in the experiments are implemented under the PyTorch framework. During training, the Adam optimizer was utilized with a batch size of 64, a learning rate set at 0.0005, and the training epochs configured to 300. Noisy labels were simulated from four label sources. The training set size for the PU dataset was 50, and for the KSC and WHLK datasets, it was 20. Patch sizes were set to 11 for the PU and KSC datasets, and seven for the WHLK dataset.

[1]Registered trademark.
[2]Trademarked.

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TABLE I
DETAIL INFORMATION OF PAVIA UNIVERSITY DATASET, KSC DATASET, AND THE WHU-HI-LONGKOU DATASET IN EXPERIMENT

Class	Pavia University		Kennedy Space Center		WHU-Hi-LongKou	
	Class Name	Samples	Class Name	Samples	Class Name	Samples
1	Asphalt	6631	Scrub	761	Corn	34511
2	Meadows	18649	Willow swamp	243	Cotton	8374
3	Gravel	2099	CP hammock	256	Sesame	3031
4	Trees	3064	Slash pine	252	Broad-leaf soybean	63212
5	PaintedMetalSheets	1345	Oak/Broadleaf	161	Narrow-leaf soybean	4151
6	BareSoil	5029	Hardwood	229	Rice	11854
7	Bitumen	1330	Swap	105	Water	67056
8	SelfBlockingBricks	3682	Graminoid marsh	431	Roads and houses	7124
9	Shadows	947	Spartina marsh	520	Mixed weed	5229
10			Cattail marsh	404		
11			Salt marsh	419		
12			Mud flat	503		
13			Water	927		
Total		42776		5211		204542

The weights were configured as follows: α was set to 0.8, while both β and γ were set to 0.1. In addition, noisy labels from four label sources were simulated based on the same-spectrum-different-object and different-spectrum-same-object features of hyperspectral data, following (1) and (2) in Section III-A. Two sets of upper bounds for proficiency levels were designed for each label source: $L_1 = \{30, 50, 60, 70\}$ and $L_2 = \{50, 90, 130, 170\}$. In L_1, noise was relatively low, around 35%, while in L_2, noise was higher, around 65%. The classification model's network backbone employed the state-of-the-art triple contrastive representation learning (TCRL) framework for learning hyperspectral noise [7].

C. Comparative Methods

Our method is the first to address the issue of multisource noise learning in HSI classification. The existing methods lack the capability to handle this specific challenge. Therefore, we have identified related approaches, as outlined below.

1) *3D-Conv [50]:* The method introduces 3D-CNN into HSI classification, utilizing all spectral bands as input without the need for preprocessing or postprocessing. It can extract spectral and spatial features of HSI simultaneously, fully leveraging the structural characteristics of 3-D HSI data.
2) *SSRN [17]:* The method is a supervised deep learning framework that utilizes raw 3-D cubes as input data. It continuously enhances spectral features and captures spatial characteristics from HSI through spectral and spatial residual blocks.
3) *DCRN [8]:* The method proposes a novel dual-channel residual network (DCRN) to address the HSI classification problem with noisy labels. DCRN enhances training by fully leveraging useful information from noisy samples, thus minimizing the influence of noisy labels on classification accuracy.
4) *TCRL [7]:* The method proposes a novel TCRL framework, which achieves robustness in HSI classification with noisy labels through cluster-level, instance-level, and structure-level loss functions.

Among these frameworks, 3D-Conv and SSRN are commonly utilized for HSI classification tasks, while DCRN and TCRL are specifically tailored to address single-source noise learning challenges, and they are SOTA methods in HSI noise learning. Our approach follows a one-sample-multilabel setup, while the four comparative methods utilize a one-sample-one-label, which implies that labels from different sources need to be repeatedly associated with the same samples. For instance, in our experiments, we simulated four label sources, and only one sample input is required in our approach, whereas the four comparative methods need feed the same sample four times into the network in the training. This increases the resource usage significantly. To this end, we devised an MV method, which determines the most frequently occurring land cover category from the four label sources as the final label for each pixel. Subsequently, we implement the MV strategy based on the TCRL for the sake of fairness.

D. Experiments on the PU Dataset

We tested 3D-Conv, SSRN, DCRN, TCRL, MV, and our method on the PU dataset. The results are presented in Table II. It can be observed that 3D-Conv and SSRN, as traditional methods that do not consider noise, exhibit the poorest classification accuracy under multisource noise conditions. For example, using Kappa as an indicator, under the L_1 proficiency levels, they achieve 74.51% and 94.87%, respectively, while under the L_2 proficiency levels, the values drop to 43.56% and 57.09%, respectively. The classification results are illustrated in Figs. 2(b) and (c), and 3(b) and (c), where it is evident that 3D-Conv and SSRN exhibit higher misclassification rates under L_2 compared with L_1. This discrepancy arises, because the L_1 proficiency levels have fewer noisy labels, resulting in less influence on classification results. Consequently, 3D-Conv and SSRN achieve higher classification accuracy when noisy labels are not considered under L_1 proficiency levels. However, with the increase of noisy labels under L_2 proficiency levels, the guidance from noisy labels during training leads to significantly lower classification accuracy.

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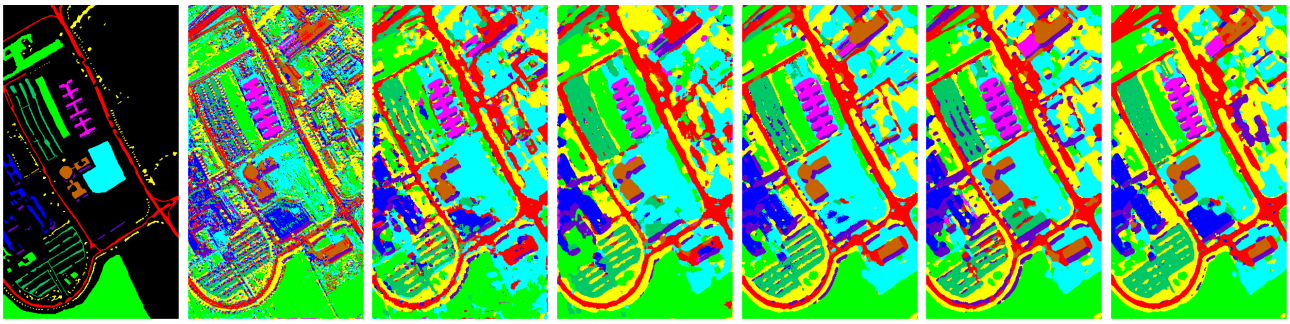
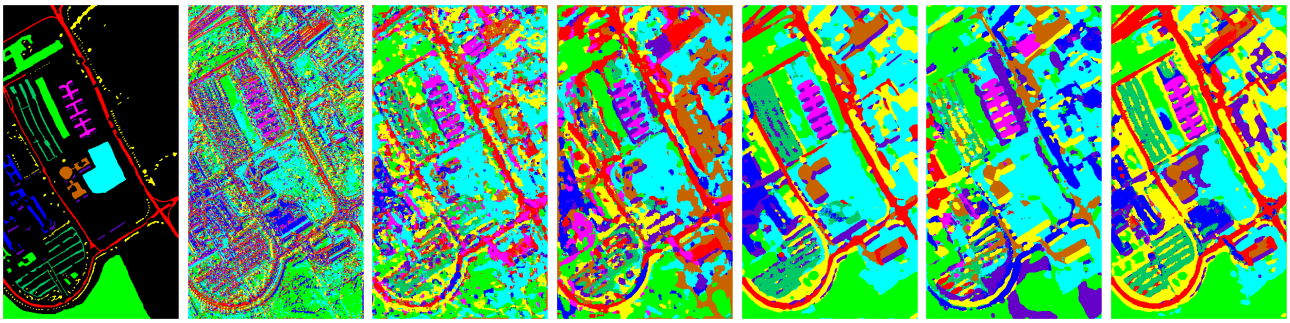


(a) (b) (c) (d) (e) (f) (g)

Fig. 2. Classification results of the six methods under L_1 proficiency levels, based on labels from four label sources in the PU dataset. (a) GT, (b) Conv-3D [50], (c) SSRN [17], (d) DCRN [8], (e) TCRL [7], (f) MV, and (g) HyperMLL.



(a) (b) (c) (d) (e) (f) (g)

Fig. 3. Classification results of the six methods under L_2 proficiency levels, based on labels from four label sources in the PU dataset. (a) GT, (b) Conv-3D [50], (c) SSRN [17], (d) DCRN [8], (e) TCRL [7], (f) MV, and (g) HyperMLL.

TABLE II

EXPERIMENTAL RESULTS ON THE DATASET PU GIVEN FOUR LABEL SOURCES AT TWO PROFICIENCY LEVELS L_1 AND L_2

Class	L_1						L_2					
	Conv-3D [50]	SSRN [17]	DCRN [8]	TCRL [7]	MV	HyperMLL	Conv-3D [50]	SSRN [17]	DCRN [8]	TCRL [7]	MV	HyperMLL
1	88.60	96.79	98.25	**99.79**	97.50	99.68	45.74	55.07	**57.91**	39.99	37.41	55.37
2	92.83	98.83	99.16	99.81	98.12	**99.92**	85.24	92.58	94.94	**98.41**	94.03	94.07
3	56.43	79.88	95.60	96.04	79.54	**98.43**	25.04	37.45	47.35	45.53	43.51	**48.86**
4	84.54	87.20	87.97	**98.26**	96.45	97.55	50.12	55.04	60.50	**72.65**	53.15	71.61
5	77.48	88.21	93.93	**99.98**	93.20	**99.98**	34.52	40.72	57.50	79.60	**90.07**	66.38
6	65.21	88.09	**97.24**	95.64	93.93	95.10	43.05	66.68	78.41	67.72	77.70	**88.52**
7	52.04	84.53	90.47	94.75	76.15	**99.70**	15.70	22.88	27.95	27.14	30.11	**38.95**
8	71.95	88.39	92.79	96.12	88.77	**99.35**	43.59	61.93	**67.78**	52.49	51.88	59.24
9	62.56	75.71	80.32	92.08	89.14	**99.77**	17.24	26.93	37.63	43.52	39.87	**46.31**
OA (%)	80.55	92.67	96.12	98.29	93.57	**98.94**	55.61	65.90	73.89	65.40	67.23	**84.30**
AA (%)	72.40	87.51	92.86	96.94	90.31	**98.83**	40.03	51.03	58.89	58.56	57.52	**63.26**
Kappa × 100	74.51	90.38	94.87	97.73	91.49	**98.60**	43.56	57.09	66.58	57.15	58.22	**78.94**

In contrast, DCRN, TCRL, and MV, as methods that consider single-source noise in HSI classification, exhibit slightly higher classification accuracy under multisource noise compared with SSRN and significantly higher accuracy than 3D-Conv. Their classification results are depicted in Figs. 2(d)–(f) and 3(b)–(f). For instance, using Kappa as an indicator, under the L_1 proficiency levels, they achieve 94.87%, 97.73%, and 91.49%, respectively, while under the L_2 proficiency levels, the values decrease to 66.58%, 57.15%, and 58.22%, respectively. However, since these three methods only consider single-source noise, their classification accuracy is still affected in the presence of multisource noise.

Our method achieves the highest class-specific classification average accuracy (AA) and the most numerous highest number of correct classifications, both under L_1 and L_2 proficiency levels. The overall accuracy (OA), AA, and Kappa values are also the highest. The classification results are shown in Figs. 2(g) and 3(g), while the GT is depicted in Figs. 2(a) and 3(a). Our classification results closely match the GT. Specifically, using Kappa as an indicator, under L_1 and L_2, they are 98.60% and 78.94%, respectively. In comparison with 3D-Conv, SSRN, DCRN, TCRL, and MV, our method achieves higher Kappa values by 32.33%, 9.09%, 3.93%, 0.89%, and 7.77%, respectively, under L_1, and by 81.22%, 38.27%, 18.56%, 38.13%, and 35.59%, respectively, under L_2. This indicates that our method not only performs well when the noise level is low but also outperforms others significantly when the noise level is high. This is because our method learns three CMs for each noise source, thereby reducing the impact of noisy labels on the classification network and improving classification accuracy.

E. Experiments on the KSC Dataset

The classification accuracies of Conv-3D, SSRN, DCRN, TCRL, MV, and our method on the KSC dataset with noisy

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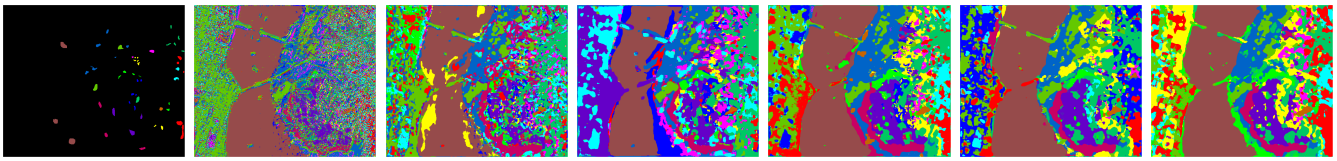
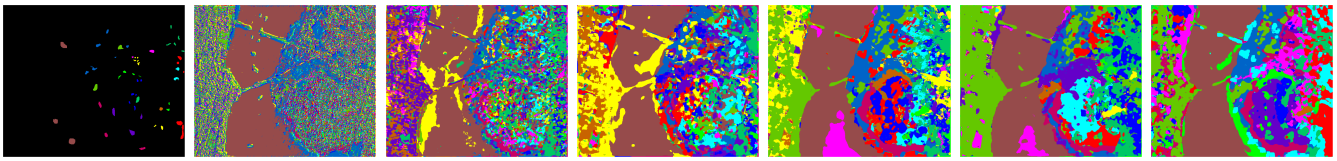


(a) (b) (c) (d) (e) (f) (g)

Fig. 4. Classification results of the six methods under L_1 proficiency levels, based on labels from four label sources in the KSC dataset. (a) GT, (b) Conv-3D [50], (c) SSRN [17], (d) DCRN [8], (e) TCRL [7], (f) MV, and (g) HyperMLL.



(a) (b) (c) (d) (e) (f) (g)

Fig. 5. Classification results of the six methods under L_2 proficiency levels, based on labels from four label sources in the KSC dataset. (a) GT, (b) Conv-3D [50], (c) SSRN [17], (d) DCRN [8], (e) TCRL [7], (f) MV, and (g) HyperMLL.

TABLE III

EXPERIMENTAL RESULTS ON THE DATASET KSC GIVEN FOUR LABEL SOURCES AT TWO PROFICIENCY LEVELS L_1 AND L_2

Class	L_1						L_2					
	Conv-3D [50]	SSRN [17]	DCRN [8]	TCRL [7]	MV	HyperMLL	Conv-3D [50]	SSRN [17]	DCRN [8]	TCRL [7]	MV	HyperMLL
1	71.39	76.47	78.32	79.74	78.63	**100.00**	13.90	11.34	14.41	8.27	0.00	**71.89**
2	49.75	64.53	71.13	76.45	75.36	**100.00**	9.31	7.51	17.38	1.52	22.22	**86.45**
3	41.97	62.63	72.43	73.68	39.48	**80.00**	3.18	0.77	0.00	5.52	0.00	**49.42**
4	30.47	**69.06**	70.31	74.76	53.31	68.68	5.64	0.53	3.94	4.00	1.62	**33.60**
5	28.80	45.73	56.95	71.74	35.98	**85.27**	1.84	0.59	0.00	0.00	0.00	**40.71**
6	31.29	55.41	68.85	65.43	36.43	**99.90**	1.88	0.00	0.00	0.00	0.00	**55.46**
7	27.68	50.61	68.05	67.82	48.96	**95.12**	0.00	0.00	0.00	0.00	0.00	**60.00**
8	65.78	79.33	80.54	94.32	87.16	**97.50**	33.06	59.39	57.48	**77.27**	71.74	62.54
9	72.72	84.20	92.19	97.30	96.74	**100.00**	25.09	38.48	42.24	61.07	30.85	**70.42**
10	79.85	87.27	94.22	99.35	95.48	**100.00**	46.44	60.02	68.98	**89.15**	81.12	58.24
11	76.07	83.56	95.76	**99.17**	95.72	97.90	47.06	47.77	59.31	82.97	72.98	**88.95**
12	75.42	88.27	89.04	90.35	95.29	**100.00**	53.48	65.21	67.70	80.59	82.88	**90.34**
13	91.65	94.36	94.34	97.28	96.58	**100.00**	78.66	85.44	83.56	**97.31**	82.68	91.75
OA (%)	69.69	80.75	86.20	89.10	82.04	**97.59**	40.27	47.03	50.10	46.61	44.11	**79.75**
AA (%)	57.14	72.42	79.39	83.65	71.93	**94.18**	24.58	29.00	31.92	39.05	34.31	**66.14**
Kappa × 100	66.21	78.55	84.64	87.93	80.09	**97.31**	33.62	41.27	44.65	41.64	38.62	**77.33**

labels from four sources are shown in Table III. It can be observed that the overall trend is consistent with that on the PU dataset, where our method achieves the highest accuracy, followed by TCRL and DCRN methods considering single-source noise, then our proposed MV method, and the lowest are SSRN and Conv-3D methods without noise consideration. Specifically, in terms of per-class AA, our method achieves the highest accuracy most frequently. With a total of 13 classes, under the L_1 proficiency levels, our method achieves the highest per-class AA 11×, with the AA of six classes reaching 100%, and one class reaching 99.9%; under the L_2 proficiency levels, our method achieves the highest per-class AA ten times. Furthermore, it can be noted that Conv-3D, SSRN, DCRN, TCRL, and MV methods display numerous instances where no correct identifications were made for land cover classes 1, 3, 5, 6, and 7. Furthermore, even for class 7, the AA of the aforementioned five methods is exceedingly low. This outcome may stem from our innovative noise simulation method, designed to emulate the occurrence of noisy labels arising from same-spectrum-different-object and different-spectrum-same-object characteristics in HSI. In our noise simulation, for each label source, we randomly select a "weak" class, indicating that this label source struggles to correctly identify this class. Then, we randomly choose a "weak" spectral vector from this class, as the representative of the spectral vectors for the "weak" class. Then, compute the spectral similarity between the "weak" spectral vector and other pixel spectral vectors. Pixels with spectral similarity less than their proficiency level are labeled as the same class as the "weak" class. It is possible that when simulating noise, we selected "weak" spectral vectors similar to classes 1, 3, 5, 6, and 7, resulting in almost all labels for these classes being incorrect, leading to an AA of 0% for these classes. The classification results of the six methods based on the KSC dataset with multisource noisy labels under L_1 and L_2 proficiency levels are shown in Figs. 4 and 5, respectively. It can be observed that the classification results of our method [e.g., Figs. 4(g) and 5(g)] are more clear and distinct, approaching the GT [shown in Figs. 4(a) and 5(a)] compared with the results of other methods [shown in Figs. 4(b)–(f) and 5(b)–(f)].

Our method outperforms the other methods in terms of OA, AA, and Kappa coefficient, as shown in Table III. Using Kappa as an example, our method achieves a Kappa coefficient of 97.31% under the L_1 proficiency levels, which is higher than Conv-3D, SSRN, DCRN, TCRL, and MV methods by 46.97%, 23.88%, 14.97%, 10.67%, and 21.50%, respectively. Under the L_2 proficiency levels, our method achieves a Kappa coefficient of 77.33%, surpassing Conv-3D, SSRN, DCRN, TCRL, and MV methods by 130.01%, 87.38%, 73.19%, 85.71%, and 100.23%, respectively. This is because our method has the

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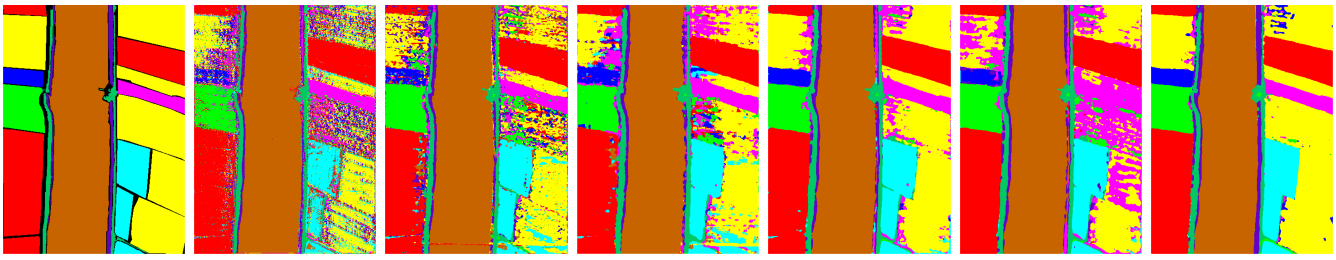
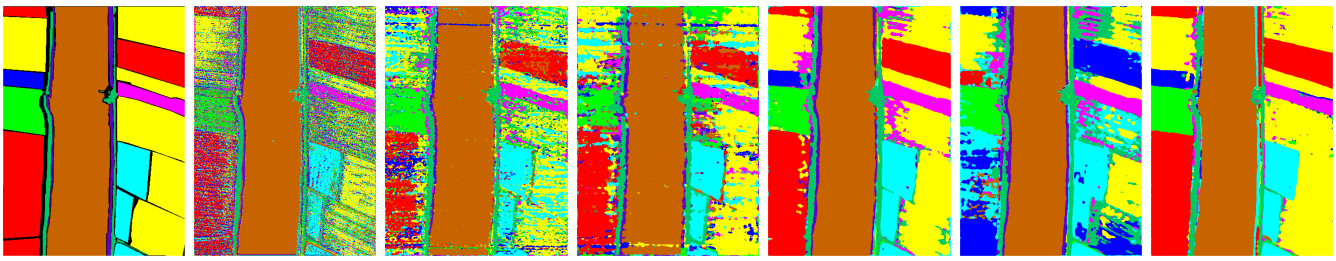


(a) (b) (c) (d) (e) (f) (g)

Fig. 6. Classification results of the six methods under L_1 proficiency levels, based on labels from four label sources in the WHLK dataset. (a) GT, (b) Conv-3D [50], (c) SSRN [17], (d) DCRN [8], (e) TCRL [7], (f) MV, and (g) HyperMLL.



(a) (b) (c) (d) (e) (f) (g)

Fig. 7. Classification results of the six methods under L_2 proficiency levels, based on labels from four label sources in the WHLK dataset. (a) GT, (b) Conv-3D [50], (c) SSRN [17], (d) DCRN [8], (e) TCRL [7], (f) MV, and (g) HyperMLL.

capability to learn the CMs of each label source, thus reducing negative effects of noisy labels on the classification network. Consequently, our method demonstrates robust performance in both low-noise (L_1) and high-noise (L_2) environments.

F. Experiments on the WHLK Dataset

In addition to the PU and KSC datasets, we also conducted simulation experiments on the high-spatial-resolution hyperspectral dataset WHLK. The classification results at proficiency levels L_1 and L_2 are illustrated in Figs. 6 and 7, respectively. It is evident that our method achieves the most accurate classification with the clearest boundaries [as shown in Figs. 6(g) and 7(g)], closely resembling the GT [as shown in Figs. 6(a) and 7(a)]. Following our method, TCRL [as shown in Figs. 6(e) and 7(e)] and DCRN [as shown in Figs. 6(d) and 7(d)], which consider noise, perform relatively well. Next is our proposed MV method [as shown in Figs. 6(f) and 7(f)], while SSRN [as shown in Figs. 6(c) and 7(c)] and Conv-3D [as shown in Figs. 6(b) and 7(b)], which do not consider noise, exhibit poorer performance. Specific per-class AA and classification precision are presented in Table IV. Notably, our method achieves the highest frequency of per-class AA at both L_1 and L_2 proficiency levels. Under L_1, our method achieves the highest per-class AA eight times out of nine class categories in the WHLK dataset, with classes 1, 4, 6, and 7 surpassing 99.50% AA. Under L_2, our method achieves the highest per-class AA six times, with class 7 reaching 98.31% AA.

Our method achieves specific accuracies of OA, AA, and Kappa coefficient on the HSI based on the WHLK dataset with noisy labels from four label sources at proficiency levels L_1 and L_2, as detailed in Table IV. While AA represents the average accuracy across all classes, OA denotes the overall proportion of correctly classified instances by the classifier across all classes, and Kappa coefficient measures the agreement between the classifier's predictions and the GT, considering the effect of random chance. It is evident that our method outperforms the other five methods in terms of classification accuracy, whether at L_1 or L_2 proficiency levels. Taking Kappa coefficient as an example, under L_1, our method achieves a Kappa coefficient of 97.96%, which is higher than Conv-3D, SSRN, DCRN, TCRL, and MV methods by 27.12%, 11.74%, 9.35%, 1.94%, and 5.44%, respectively. Under L_2, our method achieves a Kappa coefficient of 89.86%, surpassing Conv-3D, SSRN, DCRN, TCRL, and MV methods by 75.99%, 54.29%, 32.09%, 5.71%, and 81.43%, respectively. This indicates that our method also demonstrates robustness on high-spatial-resolution hyperspectral datasets. This is attributed to our method's ability to learn the labeling habits and proficiency levels of all four label sources, thus minimizing the negative influence of noisy labels on the classification model.

G. Source Quality Analysis

The multisource noisy labeling model's learning not only enhances the classification accuracy of HSI in multisource noisy environments but also facilitates the identification of more reliable label sources. As shown in (8) and (9) in Section III-C, by computing the weighted sum of the traces of SS feature confusion matrix A^r, spectral feature confusion matrix B^r, and spatial feature confusion matrix C^r, the overall credibility of label source r can be obtained. Similarly, by computing the weighted sum of the correct labeling probabilities for each individual land cover category, the class-specific credibility of label source r can be obtained. After normalizing the class-specific credibility of the four

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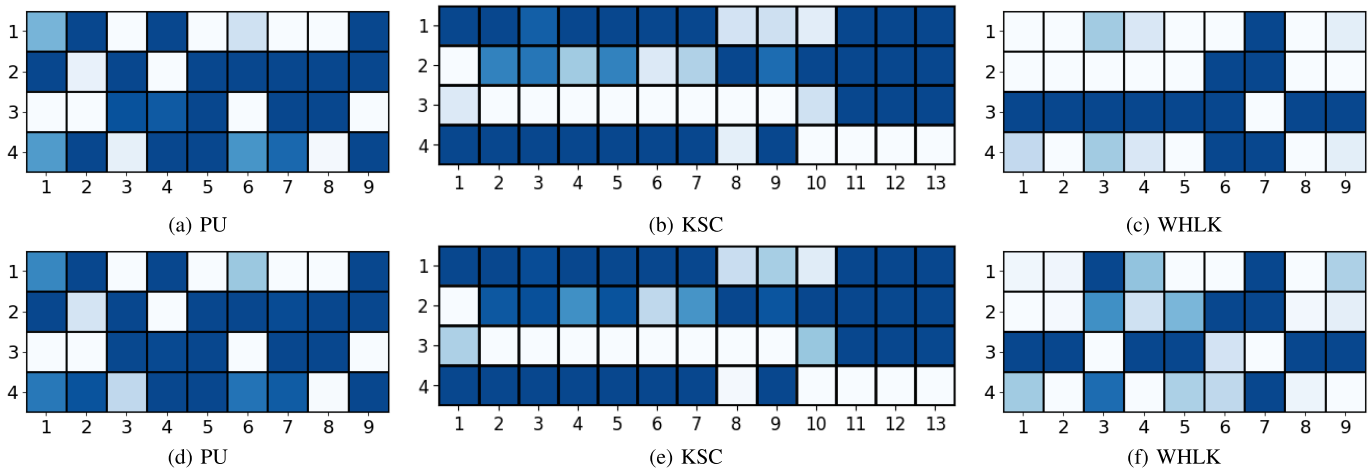
TABLE IV

EXPERIMENTAL RESULTS ON THE DATASET WHLK GIVEN FOUR LABEL SOURCES AT TWO PROFICIENCY LEVELS L_1 AND L_2

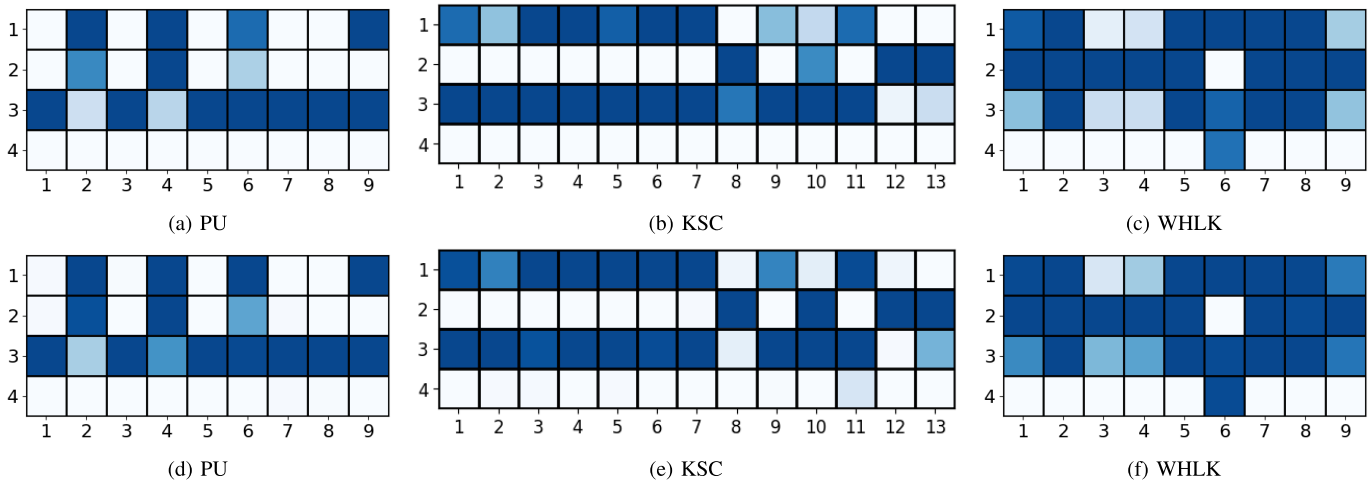
Class	L_1						L_2					
	Conv-3D [50]	SSRN [17]	DCRN [8]	TCRL [7]	MV	HyperMLL	Conv-3D [50]	SSRN [17]	DCRN [8]	TCRL [7]	MV	HyperMLL
1	83.60	93.48	91.97	99.19	99.02	**99.65**	66.58	81.00	82.28	**95.72**	75.46	92.12
2	70.57	72.85	78.52	92.96	84.58	**95.02**	34.37	44.91	51.59	78.22	67.13	**80.30**
3	25.25	48.88	72.11	91.50	**92.99**	90.01	7.39	10.66	15.90	62.59	13.63	**73.09**
4	96.20	98.19	98.91	99.48	99.69	**99.76**	74.47	81.05	88.41	**97.56**	50.51	97.07
5	38.69	57.95	67.69	71.81	63.67	**79.77**	23.19	29.57	46.09	56.17	29.35	**67.22**
6	70.58	90.14	88.23	96.68	98.55	**99.57**	37.54	46.93	51.41	85.25	78.62	**93.74**
7	97.00	98.31	99.15	99.85	99.76	**99.97**	90.28	92.74	94.90	98.10	78.76	**98.31**
8	66.45	85.37	78.16	94.97	95.43	**96.23**	36.19	44.23	53.08	72.30	41.55	**83.10**
9	53.74	58.89	68.23	82.70	78.39	**88.93**	23.43	25.41	29.88	**48.72**	19.07	47.84
OA (%)	81.86	90.40	91.88	97.00	94.38	**98.44**	60.30	66.30	74.63	88.20	57.84	**92.13**
AA (%)	66.90	78.23	82.55	92.13	90.23	**94.32**	43.71	50.72	57.06	77.18	50.45	**81.42**
Kappa × 100	77.06	87.67	89.58	96.10	92.91	**97.96**	51.06	58.24	68.03	85.01	49.53	**89.86**



(a) PU (b) KSC (c) WHLK

(d) PU (e) KSC (f) WHLK

Fig. 8. Credibility of the categories in the PU, KSC, and WHLK datasets for the four label sources at proficiency levels L_1. (a)–(c) GT. (d)–(f) Learned credibility.



(a) PU (b) KSC (c) WHLK

(d) PU (e) KSC (f) WHLK

Fig. 9. Credibility of the categories in the PU, KSC, and WHLK datasets for the four label sources at proficiency levels L_2. (a)–(c) GT. (d)–(f) Learned credibility.

label sources separately, the results obtained are shown in Figs. 8 and 9. Fig. 8 illustrates the credibility of the four label sources in multisource noisy labeling based on the PU, KSC, and WHLK datasets under the proficiency levels L_1. Fig. 8(a)–(c) represents the GT, i.e., the inherent reliability of the four label sources, while Fig. 8(d)–(f) represents the reliability learned by the model. Darker grid colors indicate more reliable label sources for the corresponding land cover category. For instance, as shown in Fig. 8(a), for land cover category 1 in the PU dataset, the grid corresponding to label source 2 is the darkest, indicating that label source 2 is the most reliable for land cover category 1's labels among the four label sources, while the grid corresponding to label source 3 is the lightest, indicating that label source 3's labels is the

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TABLE V
EFFECT OF MODULES CME AND CMA ON DATASETS PU, KSC, AND WHLK (OA (%), AA (%), AND KAPPA×100)

Dataset	Method	L_1			L_2		
		OA	AA	Kappa	OA	AA	Kappa
PU	Ours	98.94	98.83	98.60	84.30	63.26	78.94
	w/o CME	97.02	97.06	96.06	80.93	62.08	74.87
	w/o CMA	96.37	96.83	95.19	83.08	63.34	77.69
KSC	Ours	97.59	94.18	97.31	79.75	66.14	77.33
	w/o CME	92.79	82.24	91.86	78.06	63.42	75.52
	w/o CMA	92.59	81.16	91.64	78.11	62.92	75.57
WHLK	Ours	98.44	94.32	97.96	92.13	81.42	89.86
	w/o CME	91.65	84.96	89.79	85.87	68.44	82.25
	w/o CMA	93.22	85.71	91.53	85.12	65.92	81.46

least reliable for land cover category 1 in the PU dataset. A higher number of dark grids for a label source regarding high reliability of class-specific land cover categories suggests that this label source is generally more reliable. In Fig. 8(a), source 2 corresponds to the highest number of dark grids, indicating that source 2 has the highest overall credibility on the PU dataset.

The distribution of reliability learned by our model closely matches the GT, indicating that our model effectively learns the class-specific credibility and overall credibility of different label sources. Taking the KSC dataset as an example, in proficiency levels L_1, as shown in Fig. 8(b), for category 9, the grid corresponding to label source 4 is the darkest, followed by label source 2, then label source 1, and finally label source 3. This suggests that among all labels for category 9, labels from source 4 are the most reliable, followed by source 2, then source 1, with source 3 being the least reliable. The reliability order learned by our model closely matches the GT, as shown in Fig. 8(e). Under proficiency levels L_2, our method continues to perform well. Taking category 9 from the KSC dataset as an example, the GT, as shown in Fig. 9(b), indicates that label source 2 has the darkest grid color, followed by source 1, and then sources 4 and 3. The learned reliability, as shown in Fig. 9(e), matches the GT order. This demonstrates that our method effectively learns the reliability of different label sources, whether in low-noise (L_1) or high-noise (L_2) environments. However, our learned reliability of different label sources also has some shortcomings. As shown in Fig. 8(c) and (f), the learned class-specific reliability in the WHLK dataset contains errors, particularly in the evaluation of reliability for category-3 labels. In the GT [Fig. 8(c)], label source 3 is the most reliable for class 3, whereas in the learned reliability [Fig. 8(f)], label source 1 is considered the most reliable.

The assessment of class-specific credibility and overall credibility of different label sources in HSI classification research is novel in this study. Learning the credibility of different label sources not only reduces label noise for future research, such as selecting labels provided by the most reliable label source for each class as model input, but also reduces label acquisition costs. For example, in situations where label acquisition costs are limited, selecting the most reliable label sources ensures label quality while saving costs.

H. Ablation Study

We analyzed the roles of different CMs in our framework. We term the spectral feature derived CM as CME and spatial feature derived CM as CMA. We conducted ablation experiments on the CME and the CMA, with the results shown in Table V. In the PU dataset, under the L_1 proficiency levels, removing the CME module resulted in a decrease of 1.92% in OA, 1.77% in AA, and 2.54% in Kappa. The removal of the CMA module led to a decrease of 2.57% in OA, 2% in AA, and 3.14% in Kappa. Under the L_2 proficiency levels, removing the CME module resulted in a decrease of 3.37% in OA, 1.18% in AA, and 4.07% in Kappa. The removal of the CMA module led to a decrease of 1.22% in OA and 1.25% in Kappa. In the KSC dataset, under the L_1 proficiency levels, removing the CME module resulted in a decrease of 4.8% in OA, 11.94% in AA, and 5.45% in Kappa. The removal of the CMA module led to a decrease of 5% in OA, 13.02% in AA, and 5.67% in Kappa. Under the L_2 proficiency levels, removing the CME module resulted in a decrease of 1.69% in OA, 2.72% in AA, and 1.81% in Kappa. Removing the CMA module led to a decrease of 1.64% in OA, 3.22% in AA, and 1.76% in Kappa. In the WHLK dataset, under the L_1 proficiency levels, removing the CME module resulted in a decrease of 6.79% in OA, 9.36% in AA, and 8.17% in Kappa. Removing the CMA module led to a decrease of 5.22% in OA, 8.61% in AA, and 6.43% in Kappa. Under the L_2 proficiency levels, removing the CME module resulted in a decrease of 6.26% in OA, 12.98% in AA, and 7.61% in Kappa. Removing the CMA module led to a decrease of 7.01% in OA, 15.5% in AA, and 8.4% in Kappa. The ablation experiments conducted on the PU, KSC, and WHLK datasets indicate that the CME and CMA modules significantly contribute to improving the classification accuracy of the model in HSI based on multisource noisy labels.

I. Training Cost

Our method not only achieves higher classification accuracy compared with the other five comparison methods but also requires less training time. The training time–Kappa curve

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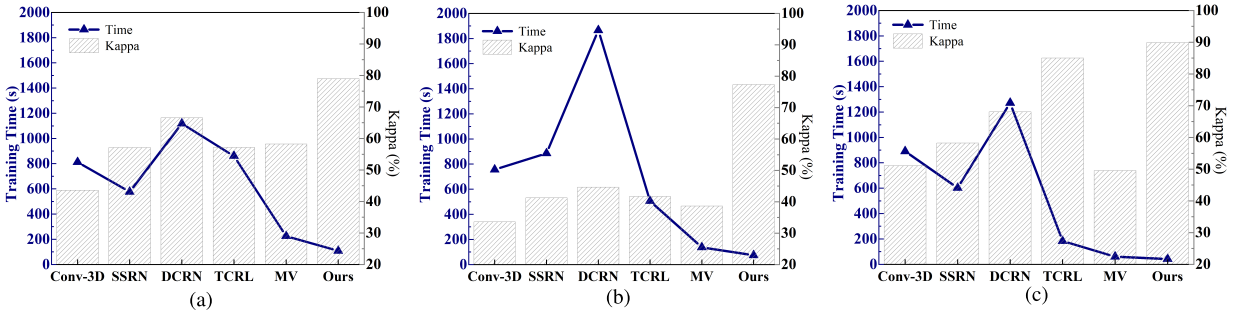

Fig. 10. Training time and Kappa curve histogram for the four label sources at the L_2 proficiency levels on (a) PU, (b) KSC, and (c) WHLK datasets.

histograms for the six methods at the L_2 proficiency levels are illustrated in Fig. 10, where (a) represents the results on the PU dataset, (b) on the KSC dataset, and (c) on the WHLK dataset. It can be observed that our method consistently achieves the highest Kappa coefficient and the lowest training time across all three datasets. This can be attributed to the fact that the Conv-3D, SSRN, DCRN, and TCRL methods are designed under the single-sample-single-label setting, whereas our method is designed under the single-sample-multilabel setting. To make these methods compatible with our setting, we replicate each sample for every label, meaning that if we simulate labels from four noise sources, each noise source corresponds to the same redundant sample. As a result, the training time and memory consumption increase significantly. In addition, we propose the MV method, which aggregates labels from four noise sources using MV to determine the label for each sample. Despite applying the state-of-the-art HSI noise learning method TCRL, the results still do not match those of our method, indicating that our method achieves the highest accuracy and shortest training time for HSI classification problems with multiple noise sources.

V. Conclusion

This article proposed a framework HyperMLL for multisource noisy learning in HSI classification. The framework not only addressed the shortcomings of multisource noisy learning in the direction of HSI classification but also introduced a new noise simulation method that better aligned with the actual conditions of HSI classification. Specifically, based on the characteristics of "same-object-different-spectral" and "different-object-same-spectral" in HSI, we developed a multisource label simulation method based on spectral similarity. Furthermore, to mitigate the negative impact of noisy labels on the classification model, we learned three CMs, which are related to spectral–spatial feature, spectral feature, and spatial feature. In addition, the learned CMs could effectively assess the overall credibility and class-specific credibility of each noise source.

The proposed framework can be improved from several aspects. We assumed that each label source would contain labels for all samples, without considering the scenario where only parts of samples were labeled in different sources. As we set hyperparameters empirically, fine-tuning these parameters might be needed for different datasets to achieve better performance. Finally, our proposed HyperMLL has not been tested on real-world multisource datasets. In future research, we will attempt to train models using noisy labels from historical annotations, remote sensing products, and large model derived labels. In addition, applications related to geoscience often involve different modalities of remote sensing data, such as HSI, SAR, and LiDAR. Therefore, multisource and cross-modal label collection and model training are also directions worth exploring in the future.

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